DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

A DPO-based method modeling TA-LLM interactions as an MDP with 5 states and 3 query types

ML Research / NLP Engineering

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Failure Mode 1

Failing to Ask Follow-up Questions

TA-LLMs with incomplete query arguments (missing slots) do not ask clarifying questions, leading to hallucinated tool calls.

Failure Mode 2

Making Hallucinated Tool Invocations

Models invoke tools with missing or incorrect arguments instead of gathering required information through dialogue.

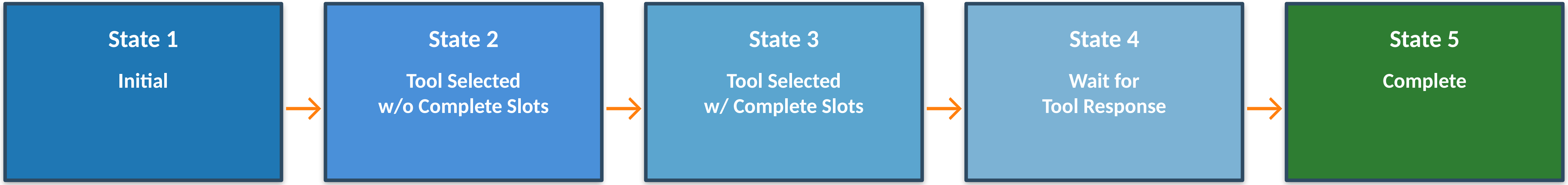
Failure Mode 3

Failing to Reject Inappropriate Tool Calls

Out-of-scope requests are not rejected properly, resulting in incorrect tool selections or responses.

Need: Preference optimization to learn correct dialogue flow beyond SFT expert trajectories

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



No history. Out-of-scope requests return rejection and remain here.

Tool identified but required parameters missing. Slot-filling QA occurs here.

Both tool selection and all argument values determined.

Tool call executed, awaiting results.

Tool results received, completion message generated.

MDP formulation enables systematic construction of state trajectories for each query type

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	1 → 3 → 4 → 5	Sufficient info provided	Query with all required arguments; proceeds directly to tool call
Type 2	1 → (2×N) → 3 → 4 → 5	Incomplete info, slot-filling needed	Query missing arguments; N rounds of follow-up Q&A until complete
Type 3	1 → 1	Out-of-scope request	Requested functionality unavailable; tool call must be rejected

Each trajectory represents a distinct dialogue pattern requiring different policy behaviors

Automatic Paired Trajectory Construction Without Human Labels

Chosen Trajectory	Correct dialogue flow (good policy)
Rejected Trajectory	Incorrect dialogue flow (bad policy)
Learning Objective	Teach specific dialogue skills via DPO

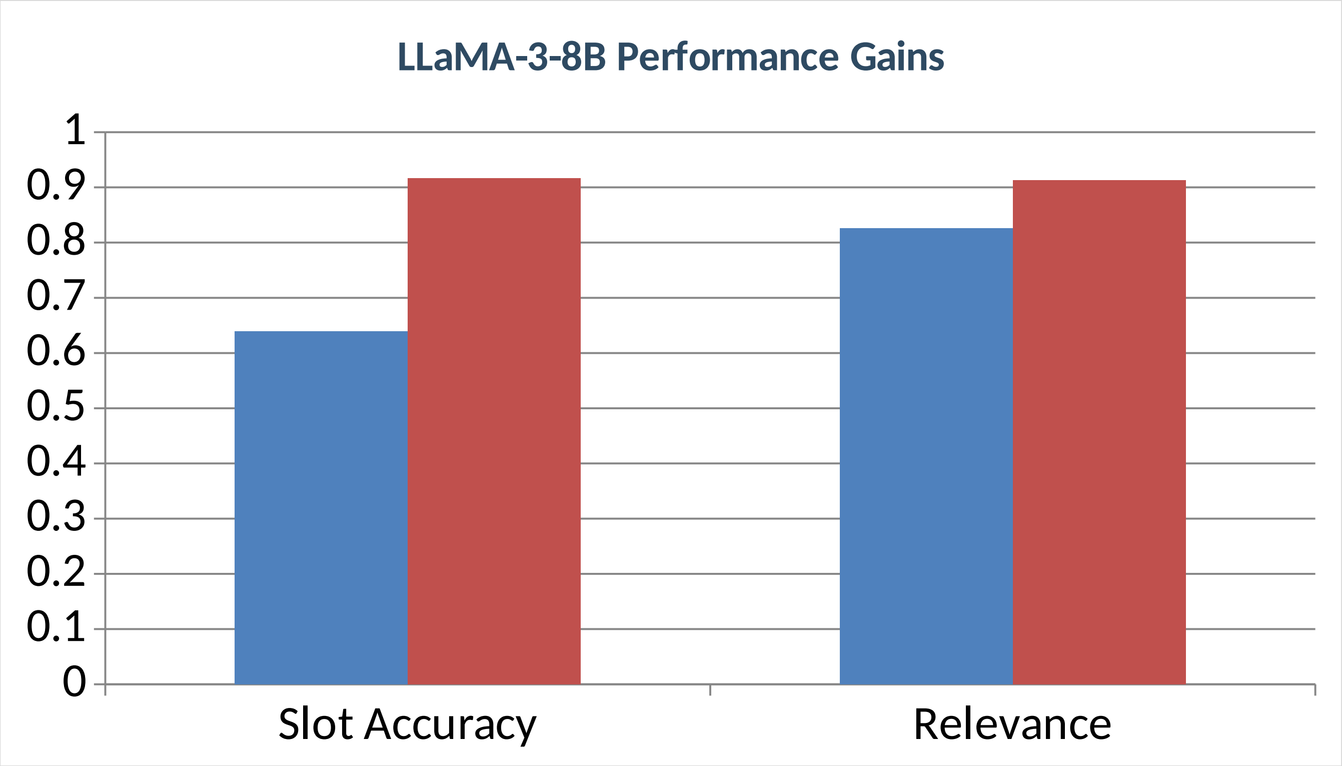
Query	Trajectory Pair	Learning Lesson	Count
Type 1	1→3→4→5 vs 1→2→3→4→5	Prevent redundant slot-filling	2,089 (Easy)
Type 1	1→3→4→5 vs 1→(2N)→3→4→5	Prevent redundant slot-filling	562 (Hard)
Type 2	1→2→3→4→5 vs 1→3→4→5	Prevent slot hallucination	2,089 (Easy)
Type 2	1→(2N)→3→4→5 vs 1→3→4→5	Prevent slot hallucination	562 (Hard)
Type 2	1→(2N)→3→4→5 vs 1→(2M)→3→4→5	Prevent slot hallucination	2,530 (Hard)
Type 3	1→1 vs 1→3→4	Tool call reject	567 (Hard)
Type 3	1→1 vs 1→(2N)→3→4	Tool call reject	562 (Hard)
		Total paired trajectories:	**8,961**

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	**0.856**
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	**0.791**
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	**0.816**
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.904

SFT + DiaTool-DPO consistently achieves the highest Macro Average across all model sizes

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Slot Accuracy

0.639 → 0.917

+43.5%

Relevance

0.826 → 0.913

+10.5%

Approach reaches near GPT-4o performance (94.8% information gathering, 91% tool call rejection)

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1

MDP formulation enables systematic trajectory pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

2

Automatic dataset construction eliminates human labeling

DiaTool-DPO generates 8,961 paired trajectories by modeling correct vs. incorrect dialogue flows.

3

DPO complements SFT for dialogue control

DiaTool-DPO-only performs poorly; DPO enhances rather than replaces supervised training.

4

Language-agnostic approach

Primary experiments in Korean generalize to English, enabling broad applicability for production TA-LLMs.

5

Path to GPT-4o-level conversational tool use on smaller models

LLaMA-3-8B with SFT + DiaTool-DPO achieves near state-of-the-art performance.